

Chapter 51 — Pathophysiology of Urinary Tract Obstruction

PATHOPHYSIOLOGY OF PAEDIATRIC URINARY TRACT OBSTRUCTION

- **Definition:** any **anatomic or functional abnormality that impedes urine flow**, injuring the kidney and lower urinary tract; may occur **anywhere from calyces to urethral meatus**, upper and/or lower tract.
- **Four classifying axes:** acute vs chronic · partial vs complete · intrinsic vs extrinsic · unilateral vs bilateral.
- The governing risk is **permanent damage to the developing kidney**.

Epidemiology

- **Overall incidence** ≈ 1 in 500 live births worldwide.
- **Fetal lower urinary tract obstruction (LUTO):** ≈ 2–3 per 10,000 fetuses (per-pregnancy figure under Urinary Tract Sites).
- Prevalence **varies by population and region**; severity forms a **spectrum**.

Association With End-Stage Renal Disease and Kidney Transplantation

- Untreated severe obstruction → **chronic kidney disease (CKD)** → **end-stage renal disease (ESRD)**.
- **High-grade obstruction:** lifetime ESRD risk up to 30%.
- ESRD means **lifelong dialysis or transplantation**, each with significant morbidity — hence the urgency of **early detection**.

KEY POINTS

- Urinary tract obstruction is **relatively common** in children.
- Severe uncorrected obstruction → **CKD, ESRD or renal failure**.

Concept — *paediatric obstruction strikes a **kidney that is still developing**: the same pressure that a mature kidney partly tolerates can permanently cap nephron number, so the subject is organised around **detecting early and decompressing before development is lost**.*

CLINICAL PRESENTATIONS OF OBSTRUCTION

- Presentation depends on **site, severity, developmental stage at onset and duration**.

Timing of Presentation

Prenatal Versus Postnatal Presentation

	Prenatal	Postnatal
Detection	Routine anomaly ultrasound (US)	Symptoms, signs or incidental imaging
Timing	From 11–14 weeks ; most at 18–20 weeks (second trimester)	Any age
Commonest finding	Hydronephrosis (collecting-system dilation)	Variable — asymptomatic to life-threatening

	Prenatal	Postnatal
Key measure	Renal pelvis anteroposterior diameter (APD) within the urinary tract dilation (UTD) classification — the most reliable parameter	—

- **Prenatal US signs** (any combination): dilated pelvis, ureters, bladder (**megacystis**) or posterior urethra (**keyhole sign**); **cystic change or increased parenchymal echogenicity**; **bladder wall thickening**; **ascites**; **oligohydramnios or anhydramnios**. Severe LUTO also **impairs pulmonary and limb development**.
- **Every structure proximal to the obstruction dilates**, so the pattern of dilation localises the level.
- Prenatal hydronephrosis **warrants further evaluation**; early detection permits **fetal intervention or planned postnatal management**.
- **Postnatal manifestations**:
 - **Mass**: abdominal distension, palpable mass
 - **Infection or pain**: unexplained fever, **recurrent urinary tract infection (UTI)**, flank or abdominal pain
 - **Urine output and voiding**: polyuria, urinary retention
 - **Systemic**: emesis, early satiety, anorexia, **failure to thrive**; fluid and electrolyte imbalance with renal dysfunction or failure
 - **Respiratory**: compromise or failure from **pulmonary hypoplasia**

Urinary Tract Sites

LOWER VERSUS UPPER URINARY TRACT OBSTRUCTION

	Lower urinary tract obstruction (LUTO)	Upper urinary tract obstruction
Frequency	1 in 5000 – 1 in 25,000 pregnancies; among the commonest genitourinary congenital anomalies	Ureteropelvic junction obstruction (UPJO): 1 in 1500 live births
Commonest cause	Posterior urethral valves (PUV) and urethral atresia ≈ 50%	UPJO — the vast majority ; male : female 2 : 1
Other causes	Anterior urethral valves, megalourethra, prolapsing ureterocele, cloacal malformation, megacystis–microcolon–intestinal hypoperistalsis syndrome (MMIHS), trisomy 18 and 21	Ureterovesical junction obstruction (UVJO), pyelocalyceal diverticulum, ureterocele, ectopic ureter, fibroepithelial polyp, stones
Imaging	Megacystis, bilateral dilation, keyhole sign	Hydronephrosis, hydroureteronephrosis or isolated calyceal dilation, by level
Outcome driver	ESRD and pulmonary hypoplasia	Progressive loss of renal function if untreated

- **LUTO mechanism**: obstructed flow and **raised intrarenal pressure compromise glomerular development**; **loss of amniotic fluid causes pulmonary hypoplasia**, raising fetal, perinatal and postnatal mortality.
- **Oligo- or anhydramnios in the second trimester → combined mortality 60–80%**.
- At both sites severity ranges from **insignificant to progressive renal deterioration**, so urgency of treatment varies.
- **Fig. 51.1** — female external genitalia with **cloacal malformation and pseudophallus**.

TABLE 51.1 PAEDIATRIC URINARY TRACT OBSTRUCTION AETIOLOGIES

Mechanism	Upper tract (kidney, ureter)	Lower tract (bladder, prostate, urethra)	Extrinsic
Congenital / structural	UPJO , obstructing megaureter, ectopic ureter, ureterocele, calyceal diverticulum (narrow infundibulum), ureteral stricture	Posterior and anterior urethral valves , urethral atresia, caecoureterocele, prostatic utricle, syringocele, urethral stricture, meatal stenosis, phimosis	Lower pole crossing vessel (UPJO)

Mechanism	Upper tract (kidney, ureter)	Lower tract (bladder, prostate, urethra)	Extrinsic
Intraluminal	Calculi, sloughed papillae (pyelonephritis, acute tubular necrosis), bleeding (infection, trauma)	Calculi, clot retention	Faecal impaction
Neoplastic	Tumour, fibroepithelial polyp	Prostatic rhabdomyosarcoma	Pelvic and spinal tumours , retroperitoneal lymphadenopathy
Functional / fibrotic	—	Neurogenic bladder, myogenic failure; detrusor hypertrophy → secondary UVJO	Retroperitoneal fibrosis

- **Purpose:** obstruction is **not only congenital** — intraluminal, neoplastic, functional and extrinsic causes apply at every level.
- **Pattern:** calculi appear at **both** levels; the crossing vessel is filed as **extrinsic** although it produces UPJO.
- **Traps:** **detrusor hypertrophy is a lower tract cause of an upper tract (UVJ) obstruction** — the thickened wall narrows the intramural ureter; **neurogenic bladder obstructs without an anatomical lesion.**

KEY POINTS

- Manifestations depend on **location, severity, age at onset and duration; location dictates the symptoms.**
- Prenatal detection is by **routine US**; postnatal presentation is **more variable**, from asymptomatic to life-threatening.
- **LUTO carries significant long-term morbidity and mortality; UPJO accounts for the vast majority of upper tract obstruction.**

Concept — LUTO is a **bilateral, whole-fetus disease** (kidneys and lungs fail together through loss of amniotic fluid), whereas upper tract obstruction is usually a **unilateral, organ-level problem** with a normal contralateral kidney — which is why LUTO dominates mortality despite being far rarer than UPJO.

DEFINITION OF OBSTRUCTION

- Obstruction is an **impedance of urinary flow** at any level, involving one or several components.

Partial	High-grade / complete
Fewer signs and symptoms; may resolve spontaneously	More signs and symptoms; requires intervention

- **Causes:** congenital malformation, structural abnormality, inflammation, tumour, extrinsic compression (Table 51.1).
- **Spectrum:** incidental imaging finding → **recurrent UTI, retention, hydronephrosis, impaired renal function.**
- **Consequences:** untreated severe obstruction → **CKD**; when prenatal, also **pulmonary hypoplasia and limb deformity.**
- **Management aims:** relieve severe obstruction → preserve renal function → prevent long-term complications, by **conservative, pharmacological, endoscopic or surgical** means chosen by cause, severity and clinical status.

KEY POINTS

- Obstruction may be **partial or high grade**, at any level, **intrinsic or extrinsic, congenital or acquired.**

- Treatment aims to **relieve obstruction, preserve renal function and prevent complications.**

Concept — *obstruction is defined by its effect on flow and function, not by dilation: a dilated system can be unobstructed and resolve alone, which is why the partial/high-grade distinction, not the image, decides treatment.*

DIAGNOSIS OF URINARY TRACT OBSTRUCTION

- **The central task:** separate children who **need surgery to preserve renal function** from those who will **resolve spontaneously.**
- Imaging and laboratory studies grade severity and time treatment.

Timing of Obstruction: Prenatal Diagnostic Testing

- **Congenital anomalies of the kidney and urinary tract (CAKUT)** are among the commonest prenatal US anomalies: **1 in 250 – 1 in 1000 pregnancies.**
- Severity is judged by **degree of hydronephrosis, associated urinary tract changes and amniotic fluid volume**; fetal blood and urine biomarkers and proteomics may add prognostic information.

Imaging

Ultrasound. The primary prenatal modality — **cost-effective, widely available, part of standard anomaly screening**; grades obstruction by **APD**, calyceal dilation, parenchymal thickness and appearance, and bladder and ureteral abnormalities.

- **Landmarks:** kidneys visible from **9 weeks**; bladder filling by **12 weeks** — **not seen by 15 weeks is abnormal; normal fetal ureters are not visualised**, so a visible ureter is dilated.
- **UTD classification:** unified prenatal and postnatal grading from **eight societies** (Society for Pediatric Urology, Society for Fetal Urology, American Society of Pediatric Nephrology, Society for Maternal-Fetal Medicine, American College of Radiology, American Institute of Ultrasound in Medicine, Society for Pediatric Radiology, Society of Radiologists in Ultrasound), stratified **before and after 28 weeks.**
- **European Rare Kidney Disease Reference Network (ERKNet) consensus, May 2022:** **APD is the most reliable parameter**; obstructive uropathy is considered at **APD ≥ 7 mm at 16–27 weeks and ≥ 10 mm at ≥ 28 weeks.**
- **Below these thresholds, obstruction is not considered** unless there is **megacystis and/or reduced amniotic fluid.**

Magnetic resonance imaging. Fetal MRI is a **selective adjunct** to US (Fig. 51.2).

- **Indications:** US technically limited (**oligo- or anhydramnios, maternal obesity**); **complex anatomy** in severe obstruction; **planning fetal intervention.**
- **Advantages:** **not limited by fetal position**; multiplanar. Three-dimensional reconstruction and functional renal imaging are developing.
- **Fig. 51.2** — T2-weighted fetal MRI of **cloacal malformation**: dilated ventral bladder, with **uterus didelphys and massive hydrometrocolpos** behind it.

Laboratory Studies

Bladder taps. Fetal vesicocentesis — US-guided aspiration of fetal urine to **distinguish obstructive from non-obstructive hydronephrosis** and **predict postnatal renal function**; tests **sodium, chloride and calcium** (most important), **osmolarity** and low-molecular-weight proteins (**β_2 -microglobulin**, proteomics). Supporting evidence is still accruing.

Urinary markers. Utility **debated**. Used to separate obstructive from non-obstructive hydronephrosis and megacystis, predict renal function and **monitor fetal intervention for LUTO**.

- **Megacystis:** enlarged bladder **failing to empty over 40 minutes of observation at ≥ 18 weeks**.
- **Three serial samples; the last reflects current renal function** (earlier samples are stagnant urine).

FAVOURABLE FETAL URINE BIOCHEMISTRY

Sodium	Chloride	Calcium	Osmolarity	β ₂ -microglobulin
< 100 mEq/L	< 90 mEq/L	< 8 mg/dL	< 200 mOsm/L	< 6 mg/L

- The healthy fetal tubule **reabsorbs sodium and chloride and makes hypotonic urine**, so **high values mean tubular damage**; every favourable value is a "less than".

Biomarkers. Investigational: **microRNAs (miRs) and protein profiles** discriminate severe, partial and non-obstructive hydronephrosis and may predict need for surgery. **12PUV**, a 12-peptide urinary signature in PUV-related LUTO, predicts **ESRD before 2 years** with **sensitivity 88%, specificity 95%**.

Postnatal Diagnostic Testing

Laboratory Studies

Urinalysis. The **initial study**; sequences further tests.

Finding	Indicates
Low specific gravity	Concentrating defect
Proteinuria, haematuria, casts	Filtration abnormality
Crystals	Nephrolithiasis
White blood cells	Infection or inflammation

Fractional excretion of sodium. FENa measures tubular sodium handling and separates **prerenal injury from intrinsic renal disease**: $\text{FENa} = (\text{urine Na} \times \text{serum creatinine}) / (\text{urine creatinine} \times \text{serum Na}) \times 100$, from **simultaneous** samples.

FENa < 1%	FENa > 1%
Sodium-avid state — prerenal azotaemia	Tubular damage — intrinsic disease, acute tubular necrosis

- **In obstruction:** FENa **may be low in acute obstruction** (filtration falls while tubular avidity is preserved); **established obstruction with tubular injury gives a high FENa** from salt wasting.

Glomerular filtration rate. GFR is the fundamental measure of renal function and guides treatment; the **most accurate paediatric estimate is cystatin C-based**, being **less dependent on muscle mass and age**.

Biomarkers. First-year panel — **serum creatinine, serum and urine cystatin C, neutrophil gelatinase-associated lipocalin (NGAL), kidney injury molecule-1 (KIM-1), transforming growth factor-β1 (TGF-β1), retinol-binding protein (RBP), microalbuminuria** — alone and combined is a **promising non-invasive way to identify infants needing expedited surgery**.

Imaging

POSTNATAL IMAGING TIERS

Tier	Modalities	Role
Primary	US , renal scintigraphy, voiding cystourethrogram (VCUG) , MR urography (MRU)	First-line anatomy and function
Secondary	Retrograde and antegrade pyelography	When primary imaging is inconclusive ; directs management

Tier	Modalities	Role
Tertiary	Intravenous pyelography (IVP), CT urography (CTU), pressure-flow studies	Selected cases or when others are unavailable

Ultrasound. Standard of care for initial postnatal assessment — hydronephrosis, ureteral dilation, anatomy — **without radiation**. **Fig. 51.3:** thickened bladder wall and bilateral hydroureter in PUV.

TECHNETIUM-99M RENAL SCINTIGRAPHY

	Diethylenetriaminepentaacetic acid (DTPA)	Dimercaptosuccinic acid (DMSA)	Mercaptoacetyltriglycine (MAG3)
Shows	Function, perfusion	Function, perfusion, cortical scarring	Function, perfusion plus excretion and drainage
Localises level	No	No	Yes

- All three assess **presence and severity of obstruction and resulting impairment**; **MAG3 is the drainage study**.
- **Fig. 51.4** — DMSA: reduced left function and **lower pole scarring** with grade 4 vesicoureteral reflux (VUR) and febrile UTIs. **Fig. 51.5** — MAG3: **delayed left drainage = ureteropelvic junction (UPJ) obstruction**.

Voiding cystourethrogram. Shows bladder and urethra, and ureter and collecting system when VUR is present; **gold standard for VUR**; identifies PUV and other LUTO and delineates trigone, bladder neck and urethra.

- **PUV — the leading cause of bladder outlet obstruction (BOO): dilated posterior urethra on the voiding phase.**
- **Caecoureterocele: filling defect at or near the trigone**, ± extension into bladder neck and proximal urethra.
- **VUR graded 1–5** by level of reflux and ureteric dilation and tortuosity; causes **recurrent UTI, scarring and long-term impairment**.
- **VUR and obstruction interact three ways:**
 1. **VUR secondary to obstruction** — high bladder pressure in BOO
 2. **VUR causing obstruction** — **secondary UPJ obstruction in high-grade VUR**
 3. **Coexistent** — the **refluxing, obstructing megaureter**
- **Fig. 51.6** — (A) right grade 4 and left grade 5 VUR with periureteral diverticula; (B) **trabeculated bladder, dilated posterior urethra — PUV without VUR**.

Magnetic resonance urography. **Anatomy and function in one test, without ionising radiation**; quantifies damage from signal change in **perfusion, concentration and excretion**. Used in CAKUT, hydronephrosis, malformations, complex hydroureteronephrosis and to find an **ectopic ureter in the incontinent child**; functional tracer-kinetic models are advancing.

SECONDARY IMAGING

Retrograde pyelography	Antegrade pyelography
Ureteric cannulation with an open-ended catheter through the natural orifice	Percutaneous direct injection — no natural orifice
Both: fluoroscopic contrast study of the collecting system when other imaging is inconclusive ; directs further management.	

TERTIARY IMAGING

Intravenous pyelography (IVP)	CT urography (CTU)
Excretory contrast radiography; now seldom used	Non-contrast, nephrographic and excretory phases; highly specific
Rough identification of hypo- or non-functioning kidney ; degree and site of obstruction	Detailed anatomy and function; preferred to IVP for accuracy and availability
For use where conventional imaging is unavailable ; bowel preparation improves accuracy	Higher radiation dose than IVP — low-dose and split-dose protocols in development

Pressure-Flow Studies

The Whitaker test. Invasive assessment of upper tract obstruction.

- **Indications:** (1) **equivocal** less invasive tests; (2) suspected obstruction with **poor renal function**; (3) suspected **intermittent** obstruction; (4) **gross dilation with positive diuresis renogram**.
- **Technique:** percutaneous access → **high-rate perfusion of the renal pelvis** with saline or dilute contrast → **serial renal pelvic and bladder pressures**.
- **Interpretation:** unobstructed → drains freely, **no appreciable pressure rise**; obstructed → **pressure > 12 cm H₂O** or a **continuous rise**.

Uroflowmetry. Measures **voided volume, maximum flow rate (Q_{max}) and flow curve** to diagnose lower tract dysfunction or obstruction.

KEY POINTS

- **Prenatal:** imaging (US, MRI) and laboratory studies (bladder taps, urinary markers, biomarkers).
- **Postnatal:** laboratory studies plus primary, secondary and tertiary imaging, as tabulated above.

Concept — *each postnatal study answers one question — US for anatomy, MAG3 for drainage, DMSA for scar, VCUG for the bladder outlet and reflux — invasive or irradiating tests are escalations used only when these fail to separate obstruction from dilation.*

PATHOLOGICAL CHANGES OF THE URINARY TRACT WITH OBSTRUCTION

- Driven by **mechanical stress, inflammatory responses and molecular signalling**.

Lower Urinary Tract**Pathological Changes in the Bladder**

- **BOO** → **detrusor hypertrophy** (larger fibres, more mass) and **hyperplasia** (more smooth muscle cells) → **wall thickening and reduced compliance**.
- Changes span **urothelium, suburothelium, detrusor cells, extracellular matrix (ECM) and nerves**.
- **Remodelling stimuli:** cyclic stretch → raised hydrostatic and cyclic hydrodynamic pressure and **hypoxia**.
- **Three-stage remodelling:** hypertrophy → compensation → decompensation.
- **Compliance is mediated by the ECM:** chronic inflammation releases **interleukin-6 (IL-6)**, **tumour necrosis factor-α (TNF-α)** and **monocyte chemoattractant protein-1 (MCP-1)** → ↑ **collagen and fibronectin** → **fibrosis** → **poorly compliant, hostile bladder**.
- **High-pressure bladder (> 40 cm H₂O)** → **risk of renal damage and failure**.

Concept — *bladder failure in obstruction is a **matrix disease**, not only a muscle disease: hypertrophy compensates, but inflammatory collagen removes compliance, and lost compliance — pressure above 40 cm H₂O — transmits injury to the kidneys.*

Upper Urinary Tract

IMMATURE VERSUS MATURE KIDNEY

Pathological Changes in the Immature Kidney	Pathological Changes in the Mature Kidney
Apoptosis from raised intrarenal pressure and ischaemia → remodelling and scarring	Cellular proliferation , especially tubular — compensatory
Impaired nephrogenesis and abnormal differentiation	Persistent obstruction → collagen deposition and fibrosis → scarring

Regulation of Renal Fibrosis

- Hallmark: ↑ **α-smooth muscle actin (α-SMA)** and **interstitial ECM**. Renal fibrosis is the predominant cause of function loss and the critical pathway to ESRD.

Pathway	Mechanism
Mechanotransduction	Mechanosensitive proteins and receptors convert stretch into biochemical signals → fibrosis
Inflammatory mediation	TNF-α and interleukins → immune-cell recruitment, fibrogenic activation, ECM deposition
Growth factors and cytokines	TGF-β superfamily (TGF-β, bone morphogenetic proteins) → Smad signalling → fibrogenesis; TNF-α, interleukins and angiotensin II contribute
MicroRNAs	Small non-coding RNAs that negatively regulate target mRNA ; miR-34a ↑ in tubular cells in human fibrosis and murine unilateral ureteral obstruction → epithelial-to-mesenchymal transition , suppression of anti-fibrotic proteins; possible therapeutic target

Concept — every fibrotic pathway converges on **interstitial ECM accumulation**: stretch starts it, inflammation and TGF-β/Smad amplify it, miRs tune it — which is why fibrosis, not dilation, is the final common path to ESRD.

Haemodynamic Changes With Renal Obstruction

- **Sequence**: urine production continues → **proximal pressure rises** → **net filtration pressure falls** → **reduced perfusion and ischaemic injury**.
- **Bilateral obstruction or a solitary kidney** → more profound disease.
- **GFR and renal blood flow** fall in direct relation to the high-pressure system, which compensatory hypertrophy and smooth muscle stretch perpetuate as hydronephrosis.
- **Ischaemic injury** — inflammation, tubular atrophy, interstitial fibrosis — begins within 2 hours of complete obstruction.

Effects of Obstruction on Renal Tubular Function

- Sustained obstruction → **hyposthenuria (concentrating defect)** and **impaired acidification** → **polyuria and electrolyte derangement**.

URINARY CONCENTRATION ABILITY — EARLY VERSUS PROLONGED OBSTRUCTION

Early — wasting	Prolonged — retention
Sodium, potassium, bicarbonate, magnesium, calcium, phosphate	Potassium, hydrogen, chloride, ammonium

- **Distal renal tubular acidosis** is characteristic; **potassium is wasted early and retained late**.
- **Ion transport changes**: obstruction **downregulates aquaporins (AQP1-4)** — proximal tubule, descending thin limb, collecting duct — **and sodium transporters** throughout the nephron → **concentrating defect and salt wasting**; impaired **H⁺ transport** → **metabolic acidosis** in severe cases.

Egress of Urine From the Obstructed Kidney

- Partial vs complete obstruction sets **how far egress is compromised**; retained urine → **progressive pelvicalyceal dilation (hydronephrosis)** → raised intrarenal pressure → long-term damage.

KEY POINTS

- Changes in both tracts are caused by **mechanical stress, inflammation and molecular signalling**.
- **Lower tract**: detrusor hypertrophy and hyperplasia with deleterious ECM change. **Upper tract**: impaired nephrogenesis and abnormal development.
- Chronic obstruction → **scarring, impaired tubular and concentrating function, altered ion transport**; haemodynamic change; **reduced perfusion → inflammation, tubular atrophy, interstitial fibrosis**.

Concept — the **age of the kidney decides the injury**: the immature kidney dies and fails to develop, the mature kidney proliferates then scars — and in both, falling perfusion and transporter loss explain the polyuria, acidosis and salt wasting seen clinically.

CLINICAL IMPACT OF RENAL OBSTRUCTION

Hypertension

- Obstruction causes **secondary hypertension**; renal parenchymal disease accounts for **60–80% of secondary hypertension in children**.

Bilateral ureteral obstruction

≈ 77% hypertensive

Unilateral ureteral obstruction

≈ 20% hypertensive; **renin secretion lateralises to the affected kidney** in most

- **Mechanism**: ↑ intrarenal pressure → ↓ perfusion and GFR → **renin-angiotensin-aldosterone system (RAAS) activation** → ↑ angiotensin II and aldosterone → **vasoconstriction and sodium retention**.

Compensatory Renal Growth

- Significant functional loss → **compensatory hypertrophy** driven by **insulin-like growth factor-1 (IGF-1)**, **TGF-β1** and **hepatocyte growth factor (HGF)**: glomerular enlargement and tubular hypertrophy — **glomerulomegaly, not nephrogenesis**.

Renal Recovery

Chronic renal insufficiency risk. Injury begins **within 2 hours**; prolonged obstruction → scarring and atony → **end-stage renal failure over days to months**. Treatment can reverse damage; **incomplete recovery is common**.

Acute-on-chronic renal injury. Acute obstruction on pre-existing CKD → **rapid deterioration into acute kidney injury (AKI)**.

Renal reserve. The capacity to raise function under demand — **compromised by obstruction**, limiting the response to further insults.

Postobstructive Diuresis

Unilateral obstruction

Postobstructive diuresis (POD) in ≈ 2%

Bilateral obstruction

POD in ≈ 50%

- Polyuric phase after **acute relief** of obstruction, restoring euvoelaemia and plasma composition.
- **Mechanism**: **aquaporin downregulation, loss of medullary tonicity** and **tubular transporter dysfunction from apoptosis**.

- Severe fluid and electrolyte shifts → **monitor urine output and serum and urine electrolytes closely** (see Chapter 90).

KEY POINTS

- Obstruction-related hypertension is **secondary hypertension**; renal parenchymal disease causes **60–80% of secondary hypertension in children**.
- Significant functional loss → **compensatory hypertrophy**. Prolonged or recurrent obstruction **raises CKD risk**; **POD follows acute relief** with fluid and electrolyte imbalance.

Concept — bilaterality is the multiplier: with a normal contralateral kidney, hypertension, POD and renal failure are uncommon because the healthy kidney compensates and regulates volume; remove that buffer and each complication becomes the rule.

TREATMENT OF OBSTRUCTION

- Approach set by **location and aetiology**; **multidisciplinary** because decompression carries medical complications. Aims: **relieve symptoms, preserve renal function, improve outcomes**.

Fetal Interventions

Vesicoamniotic shunt or fetal cystoscopy	Amnioinfusion
LUTO with preserved renal function	Bilateral renal agenesis, or LUTO with fetal renal failure
Treats the obstruction — fetal urine restores amniotic fluid	Does not treat the obstruction — artificially replaces fluid
Regional anaesthesia, US guidance	Serial infusions through pregnancy
Both aim to prevent pulmonary hypoplasia and other sequelae of anhydramnios.	

Pain Management

- **Sources:** baroreceptor activation by obstruction, stent discomfort, infection, **decompression haematuria**.
- **Agents:** non-steroidal anti-inflammatory drugs (NSAIDs) only if renal function is not compromised, paracetamol, **α-adrenergic blockers**, anticholinergics, occasionally opioids.

Lower Urinary Tract

Drainage

LOWER TRACT DRAINAGE — CHOSEN BY AETIOLOGY, AGE, ANATOMY AND STABILITY

Option	Indication	Key point
Timed and double voiding	Mild or functional obstruction (poor bladder neck opening without anatomical lesion or sphincter overactivity)	Void every 1–3 hours by the clock; double void for raised post-void residual
Clean intermittent catheterisation (CIC)	Neurogenic bladder, selected urethral abnormalities	Empties bladder; ↓ UTI, stones, retention
Continuous overnight catheter drainage (COCD)	Progressive renal disease with dysfunctional bladder (e.g. after PUV) and nocturnal polyuria	Night-time Foley delays renal progression ; does not replace augmentation or daytime CIC
Suprapubic tube	Urethral catheterisation not feasible or contraindicated	Reliable short-term drainage
Vesicostomy	Infants with complex LUTO; unstable patients needing prompt decompression; unable or unwilling to perform CIC	Bladder dome to lower abdomen; temporising, reversed at definitive repair

Option	Indication	Key point
Surgical intervention	Severe or complex LUTO	Urethral stricture correction, valve excision , bladder outlet reconstruction

Upper Urinary Tract

Drainage

UPPER TRACT DRAINAGE — PROMPT RELIEF TO PRESERVE FUNCTION

Option	Indication	Key point
Ureteral stent	Temporising when unstable or instruments cannot pass; or with definitive treatment (laser lithotripsy, stricture dilation, pyeloplasty)	Excellent for extrinsic ureteral obstruction
Percutaneous nephrostomy	Ureteral obstruction, hydronephrosis	Flank to renal pelvis, US-guided; needs some hydronephrosis
Cutaneous ureterostomy	Traditional bladder drainage not feasible or ineffective	End or loop , by aetiology and preference

- **Fig. 51.7** — left loop cutaneous ureterostomy with a ureteral stent in the proximal segment.

Surgical Intervention

Ureteropelvic junction — pyeloplasty	Ureterovesical junction — reimplantation
Open, laparoscopic or robot-assisted laparoscopic	Technique by anatomy and surgeon preference
Identify and correct all intrinsic and extrinsic causes to prevent recurrence	Non-refluxing neo-ureterovesical anastomosis preferred in children
Aberrant vessel → re-anastomose anterior to the vessel	Prevents postoperative VUR

Management of Oliguria

- In severe obstruction or AKI: **treat the cause, optimise fluid and electrolytes, support renal perfusion.**

Management of Postobstructive Diuresis

- **Monitor intake and output, correct electrolytes, replace fluid as needed.**
- **Usually resolves within 24 hours; seldom lasts beyond 48 hours.**
- **Do not overcorrect** — over-replacement perpetuates POD.
- **Risk factors after valve ablation: oligohydramnios** (prolonged, severe obstruction); **preterm birth** (low aquaporin-2 expression); **raised serum creatinine** (renal impairment).

Management of Chronic Kidney Disease

- **Blood pressure control, dietary modification and medication** to slow progression; severe disease or ESRD → **peritoneal dialysis, haemodialysis or transplantation.**

Emerging Therapeutic Options With Precision Medicine

- **Individualised treatment based on patient characteristics and genetics** may allow more targeted interventions.

KEY POINTS

- Treatment depends on **location, aetiology and duration** of obstruction.

- It aims to **relieve obstruction and manage disease- and treatment-related complications.**

Concept — treatment runs on a **temporise-then-define** logic: drainage buys renal function and stability, definitive reconstruction follows, and the complications of relief are managed as carefully as the obstruction itself.